

KRASNOSHCHELYE, Russian Federation

WMO: 222350

Lat: **67.3495N**Lon: **37.0529E**Elev: **157**StdP: **99.45**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.4	-29.9	-36.0	0.1	-33.4	-32.3	0.2	-29.8	8.2	-2.4	7.2	-3.7	0.7	290	0.721

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	24.9	16.9	22.5	15.6	20.4	14.6	17.9	22.7	16.6	21.2	15.2	19.2	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.6	19.9	14.5	10.5	18.7	13.2	9.6	17.3	51.1	22.7	47.1	21.2	43.1	19.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 7.2	6.2	5.4	DB	-36.6	27.2	3.8	2.1	-39.3	28.7	-41.6	30.0	-43.7	31.2	-46.5	32.7	(4)
(5)			WB	-36.7	19.4	3.8	1.7	-39.4	20.6	-41.6	21.6	-43.8	22.5	-46.5	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-12.9	-13.2	-8.5	-2.4	3.5	9.6	13.6	11.2	6.6	0.4	-6.0	-9.3	(6)
(7)		DBStd	10.71	8.16	8.36	6.05	4.82	4.58	4.41	4.08	3.66	3.47	4.25	6.06	7.23	(7)
(8)		HDD10.0	4098	708	649	574	373	208	60	12	28	112	297	480	597	(8)
(9)		HDD18.3	6897	967	882	833	623	461	263	154	224	351	555	730	855	(9)
(10)		CDD10.0	255	0	0	0	0	6	49	124	64	11	0	0	0	(10)
(11)		CDD18.3	13	0	0	0	0	0	2	8	3	0	0	0	0	(11)
(12)		CDH23.3	134	0	0	0	0	2	24	81	27	0	0	0	0	(12)
(13)		CDH26.7	18	0	0	0	0	0	3	11	4	0	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.3	2.4	2.6	2.7	2.7	2.3	2.2	2.4	2.5	2.4	2.5	(14)	
(15)	Precipitation	PrecAvg	519	30	23	28	29	40	54	64	71	52	50	44	34	(15)
(16)		PrecMax	779	65	47	55	77	79	108	138	149	101	106	88	80	(16)
(17)		PrecMin	340	10	8	8	3	17	7	1	24	0	10	16	15	(17)
(18)		PrecStd	99	14	9	12	15	16	23	34	31	24	25	18	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	1.9	4.7	10.4	21.2	26.4	28.3	26.4	18.5	10.4	5.3	2.3	(19)
(20)			MCWB	0.7	0.8	1.7	6.1	12.8	16.3	18.6	18.0	13.6	9.1	5.0	1.3	(20)
(21)		2%	DB	0.0	0.3	1.9	7.4	16.9	22.8	25.7	22.7	16.4	8.6	3.1	1.2	(21)
(22)			MCWB	-0.3	-0.3	0.4	3.9	10.4	14.7	17.4	16.1	12.5	7.6	2.7	0.6	(22)
(23)		5%	DB	-1.4	-0.9	0.6	5.4	13.9	20.3	23.5	20.0	14.0	7.3	1.8	0.3	(23)
(24)			MCWB	-1.8	-1.5	-0.6	2.6	8.8	13.4	16.5	15.0	11.3	6.5	1.4	-0.2	(24)
(25)		10%	DB	-2.8	-2.5	-0.5	3.9	11.3	17.9	21.5	17.5	12.2	6.1	0.8	-0.8	(25)
(26)			MCWB	-3.2	-3.1	-1.4	1.6	7.2	12.1	15.3	13.7	10.4	5.4	0.5	-1.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.0	2.2	6.5	13.4	17.9	20.2	19.3	14.5	9.5	5.0	1.6	(27)
(28)			MCDWB	1.1	1.5	4.1	10.5	20.1	24.9	25.3	25.0	17.5	10.1	5.2	2.1	(28)
(29)		2%	WB	-0.3	-0.1	0.6	4.2	10.9	15.7	18.6	16.9	12.9	7.9	2.8	0.7	(29)
(30)			MCDWB	0.0	0.3	1.6	6.7	15.4	20.6	23.6	21.5	15.4	8.5	3.0	1.1	(30)
(31)		5%	WB	-1.8	-1.5	-0.4	2.9	9.2	14.0	17.4	15.5	11.7	6.6	1.5	-0.1	(31)
(32)			MCDWB	-1.5	-0.9	0.4	4.9	13.4	19.5	22.1	19.2	13.7	7.2	1.7	0.3	(32)
(33)		10%	WB	-3.2	-3.0	-1.4	1.9	7.5	12.5	16.2	14.2	10.4	5.4	0.5	-1.2	(33)
(34)			MCDWB	-2.8	-2.5	-0.5	3.5	11.0	17.4	20.3	17.2	12.0	6.0	0.8	-0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	7.7	9.0	8.1	7.5	8.8	9.1	7.8	6.6	4.4	5.5	6.5	(35)
(36)			MCDBR	5.1	5.4	6.6	9.0	12.7	12.9	13.0	12.1	10.0	4.8	3.5	4.4	(36)
(37)			MCWBR	5.0	5.2	5.5	6.1	7.4	6.3	6.1	6.1	6.5	4.1	3.6	4.3	(37)
(38)		5% WB	MCDBR	5.0	5.5	5.7	7.9	11.8	11.3	11.0	10.3	8.8	4.4	3.4	4.4	(38)
(39)			MCWBR	5.0	5.3	5.0	5.6	7.2	6.2	6.0	5.8	6.2	4.0	3.6	4.3	(39)
(40)			Clear-Sky Solar Irradiance	taub	0.206	0.261	0.298	0.326	0.339	0.337	0.361	0.339	0.315	0.281	N/A	N/A
(41)	taud	1.910		2.056	2.109	2.152	2.352	2.484	2.432	2.469	2.506	2.271	N/A	N/A	(41)	
(42)	Ebn at Noon	128		542	731	818	856	870	830	813	746	532	N/A	0	(42)	
(43)	Edh at Noon	12		52	87	110	101	92	94	81	61	42	N/A	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.05	0.51	1.98	3.79	4.58	5.00	4.73	3.20	1.75	0.55	0.10	0.00	(44)	
(45)		RadStd	0.00	0.06	0.12	0.25	0.41	0.53	0.59	0.31	0.18	0.10	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.63	+2.67	+2.50	N/A	N/A	N/A	-129	-184	N/A	N/A	
(47) Regional (6 neighbors)		+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page